

# 25PY101: Engineering Physics

Dr. Sreekar Guddeti

May 21, 2026

## Viva sets

## M2T2 Review

---

### Contents

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| <b>1</b> | <b>Quantum Free Electron Theory – Set 1</b>    | <b>3</b>  |
| 1.1      | Intermediate Level – Individual Type . . . . . | 3         |
| 1.2      | Intermediate Level – Group Type . . . . .      | 3         |
| 1.3      | Advanced Level – Individual Type . . . . .     | 4         |
| 1.4      | Advanced Level – Group Type . . . . .          | 4         |
| <b>2</b> | <b>Quantum Free Electron Theory – Set 2</b>    | <b>5</b>  |
| 2.1      | Intermediate Level – Individual Type . . . . . | 5         |
| 2.2      | Intermediate Level – Group Type . . . . .      | 5         |
| 2.3      | Advanced Level – Individual Type . . . . .     | 6         |
| 2.4      | Advanced Level – Group Type . . . . .          | 6         |
| <b>3</b> | <b>Opto-electronics – Set 1</b>                | <b>7</b>  |
| 3.1      | Intermediate Level – Individual Type . . . . . | 7         |
| 3.2      | Intermediate Level – Group Type . . . . .      | 7         |
| 3.3      | Advanced Level – Individual Type . . . . .     | 8         |
| 3.4      | Advanced Level – Group Type . . . . .          | 8         |
| <b>4</b> | <b>Opto-electronics – Set 2</b>                | <b>9</b>  |
| 4.1      | Intermediate Level – Individual Type . . . . . | 9         |
| 4.2      | Intermediate Level – Group Type . . . . .      | 9         |
| 4.3      | Advanced Level – Individual Type . . . . .     | 10        |
| 4.4      | Advanced Level – Group Type . . . . .          | 10        |
| <b>5</b> | <b>Optical Fibers and Lasers – Set 1</b>       | <b>11</b> |
| 5.1      | Intermediate Level – Individual Type . . . . . | 11        |
| 5.2      | Intermediate Level – Group Type . . . . .      | 11        |
| 5.3      | Advanced Level – Individual Type . . . . .     | 12        |
| 5.4      | Advanced Level – Group Type . . . . .          | 12        |

|           |                                                |           |
|-----------|------------------------------------------------|-----------|
| <b>6</b>  | <b>Optical Fibers and Lasers – Set 2</b>       | <b>13</b> |
| 6.1       | Intermediate Level – Individual Type . . . . . | 13        |
| 6.2       | Intermediate Level – Group Type . . . . .      | 13        |
| 6.3       | Advanced Level – Individual Type . . . . .     | 14        |
| 6.4       | Advanced Level – Group Type . . . . .          | 14        |
| <b>7</b>  | <b>Quantum Mechanics Principles – Set 1</b>    | <b>15</b> |
| 7.1       | Intermediate Level – Individual Type . . . . . | 15        |
| 7.2       | Intermediate Level – Group Type . . . . .      | 15        |
| 7.3       | Advanced Level – Individual Type . . . . .     | 16        |
| 7.4       | Advanced Level – Group Type . . . . .          | 16        |
| <b>8</b>  | <b>Quantum Mechanics Principles – Set 2</b>    | <b>17</b> |
| 8.1       | Intermediate Level – Individual Type . . . . . | 17        |
| 8.2       | Intermediate Level – Group Type . . . . .      | 17        |
| 8.3       | Advanced Level – Individual Type . . . . .     | 18        |
| 8.4       | Advanced Level – Group Type . . . . .          | 18        |
| <b>9</b>  | <b>Quantum Band Theory – Set 1</b>             | <b>19</b> |
| 9.1       | Intermediate Level – Individual Type . . . . . | 19        |
| 9.2       | Intermediate Level – Group Type . . . . .      | 19        |
| 9.3       | Advanced Level – Individual Type . . . . .     | 20        |
| 9.4       | Advanced Level – Group Type . . . . .          | 20        |
| <b>10</b> | <b>Quantum Band Theory – Set 2</b>             | <b>21</b> |
| 10.1      | Intermediate Level – Individual Type . . . . . | 21        |
| 10.2      | Intermediate Level – Group Type . . . . .      | 21        |
| 10.3      | Advanced Level – Individual Type . . . . .     | 22        |
| 10.4      | Advanced Level – Group Type . . . . .          | 22        |
| <b>11</b> | <b>Answer Key</b>                              | <b>23</b> |
| 11.1      | Quantum Free Electron Theory – Set 1 . . . . . | 23        |
| 11.2      | Quantum Free Electron Theory – Set 2 . . . . . | 23        |
| 11.3      | Opto-electronics – Set 1 . . . . .             | 23        |
| 11.4      | Opto-electronics – Set 2 . . . . .             | 24        |
| 11.5      | Optical Fibers and Lasers – Set 1 . . . . .    | 24        |
| 11.6      | Optical Fibers and Lasers – Set 2 . . . . .    | 24        |
| 11.7      | Quantum Mechanics Principles – Set 1 . . . . . | 25        |
| 11.8      | Quantum Mechanics Principles – Set 2 . . . . . | 25        |
| 11.9      | Quantum Band Theory – Set 1 . . . . .          | 25        |
| 11.10     | Quantum Band Theory – Set 2 . . . . .          | 26        |

# 1 Quantum Free Electron Theory – Set 1

## 1.1 Intermediate Level – Individual Type

### 1. MCQ

The probability of occupancy of an electron state at absolute zero temperature is:

- (a) 0 for all energies
- (b) 1 for all energies
- (c) 1 below Fermi energy and 0 above it
- (d) 0 below Fermi energy and 1 above it

### 2. True/False Type

Statement 1: Electronic specific heat of metals is very small compared to lattice specific heat.

Statement 2: Only electrons near the Fermi level participate in thermal excitation.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

### 3. MCQ

Density of states represents:

- (a) Number of electrons per atom
- (b) Number of available energy states per unit energy range
- (c) Number of holes in a semiconductor
- (d) Number of phonons in a crystal

## 1.2 Intermediate Level – Group Type

### 1. MCQ

Which of the following is a success of Quantum Free Electron Theory?

- (a) Explains crystal structure completely
- (b) Explains electrical conductivity of metals
- (c) Explains ferromagnetism completely
- (d) Explains superconductivity completely

### 1.3 Advanced Level – Individual Type

#### 1. Assertion and Reason

Assertion (A): Fermi-Dirac statistics is applicable to electrons in solids.

Reason (R): Electrons obey Pauli exclusion principle.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

#### 2. Assertion and Reason

Assertion (A): Quantum free electron theory predicts lower electronic specific heat than classical theory.

Reason (R): Only electrons near the Fermi level contribute to specific heat.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 1.4 Advanced Level – Group Type

#### 1. Assertion and Reason

Assertion (A): Quantum free electron theory fails to explain why some solids are insulators.

Reason (R): The theory neglects periodic lattice potential.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 2 Quantum Free Electron Theory – Set 2

### 2.1 Intermediate Level – Individual Type

1. MCQ

At absolute zero temperature, the highest occupied energy level is called:

- (a) Valence band
- (b) Conduction band
- (c) Fermi energy
- (d) Forbidden band

2. True/False Type

Statement 1: Density of states increases with energy in a free electron model.

Statement 2: All electrons contribute equally to conduction.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. MCQ

Which distribution function is used for electrons in solids?

- (a) Maxwell-Boltzmann distribution
- (b) Bose-Einstein distribution
- (c) Fermi-Dirac distribution
- (d) Gaussian distribution

### 2.2 Intermediate Level – Group Type

1. MCQ

One major failure of quantum free electron theory is:

- (a) Explaining thermal conductivity
- (b) Explaining photoelectric effect
- (c) Explaining band gaps in solids
- (d) Explaining electrical conductivity

## 2.3 Advanced Level – Individual Type

### 1. Assertion and Reason

Assertion (A): At temperatures above 0 K, some electrons occupy states above the Fermi energy.

Reason (R): Thermal energy excites electrons to higher energy states.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 2. Assertion and Reason

Assertion (A): Density of states is important in determining electron population.

Reason (R): Electron occupation depends only on temperature.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 2.4 Advanced Level – Group Type

### 1. Assertion and Reason

Assertion (A): Quantum free electron theory successfully explains Wiedemann-Franz law.

Reason (R): Both electrical and thermal conductivities are due to free electrons.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 3 Opto-electronics – Set 1

#### 3.1 Intermediate Level – Individual Type

1. MCQ

In forward bias condition of a p-n junction diode:

- (a) Barrier potential increases
- (b) Barrier potential decreases
- (c) No current flows
- (d) Depletion region widens

2. True/False Type

Statement 1: LED emits light when forward biased.

Statement 2: LEDs are made using indirect band gap semiconductors.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. MCQ

The main energy conversion in a solar cell is:

- (a) Electrical to thermal
- (b) Thermal to mechanical
- (c) Light to electrical
- (d) Electrical to light

#### 3.2 Intermediate Level – Group Type

1. MCQ

Which semiconductor is preferred for LED fabrication?

- (a) Germanium
- (b) Silicon
- (c) Gallium Arsenide
- (d) Carbon

### 3.3 Advanced Level – Individual Type

1. **Assertion and Reason**

Assertion (A): Solar cells operate under reverse bias conditions.

Reason (R): Reverse bias widens depletion region and helps separation of carriers.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

2. **Assertion and Reason**

Assertion (A): LEDs are highly energy efficient.

Reason (R): Most of the input electrical energy is converted into light energy.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 3.4 Advanced Level – Group Type

1. **Assertion and Reason**

Assertion (A): Direct band gap semiconductors are preferred for LEDs.

Reason (R): Electron-hole recombination releases photons efficiently.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 4 Opto-electronics – Set 2

### 4.1 Intermediate Level – Individual Type

#### 1. MCQ

In reverse bias condition of a p-n junction diode:

- (a) Depletion region decreases
- (b) Barrier potential decreases
- (c) Majority carrier current increases
- (d) Depletion region increases

#### 2. True/False Type

Statement 1: Solar cells generate electricity without external bias.

Statement 2: Solar cells work on photovoltaic effect.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

#### 3. MCQ

Indirect band gap semiconductors are generally unsuitable for LEDs because:

- (a) They are highly conductive
- (b) Recombination is non-radiative
- (c) They have zero band gap
- (d) They are insulators

### 4.2 Intermediate Level – Group Type

#### 1. MCQ

Which characteristic curve is commonly used for solar cells?

- (a) V-I characteristics
- (b) B-H characteristics
- (c) Stress-strain curve
- (d) Magnetization curve

### 4.3 Advanced Level – Individual Type

#### 1. Assertion and Reason

Assertion (A): Reverse current in a diode is very small.

Reason (R): Only minority charge carriers contribute to reverse current.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

#### 2. Assertion and Reason

Assertion (A): Solar cells are preferred renewable energy devices.

Reason (R): Solar energy is abundant and non-polluting.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 4.4 Advanced Level – Group Type

#### 1. Assertion and Reason

Assertion (A): LEDs have longer life compared to incandescent lamps.

Reason (R): LEDs operate with lower heat generation.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 5 Optical Fibers and Lasers – Set 1

### 5.1 Intermediate Level – Individual Type

1. **MCQ**

Laser light is characterized by:

- (a) Incoherence
- (b) Monochromaticity
- (c) Low intensity
- (d) Random phase

2. **True/False Type**

Statement 1: Stimulated emission produces coherent light.

Statement 2: Population inversion is essential for laser action.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. **MCQ**

The working principle of optical fiber is:

- (a) Diffraction
- (b) Refraction
- (c) Total internal reflection
- (d) Polarization

### 5.2 Intermediate Level – Group Type

1. **MCQ**

Numerical aperture of optical fiber represents:

- (a) Light absorption
- (b) Light guiding ability
- (c) Fiber thickness
- (d) Signal distortion

### 5.3 Advanced Level – Individual Type

#### 1. Assertion and Reason

Assertion (A): Laser light is highly directional.

Reason (R): Stimulated emission produces photons moving in the same direction.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

#### 2. Assertion and Reason

Assertion (A): Optical fibers are widely used in communication systems.

Reason (R): Optical fibers provide high bandwidth and low transmission loss.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 5.4 Advanced Level – Group Type

#### 1. Assertion and Reason

Assertion (A): Population inversion cannot occur naturally in most laser systems.

Reason (R): Lower energy states normally contain more atoms than higher energy states.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 6 Optical Fibers and Lasers – Set 2

### 6.1 Intermediate Level – Individual Type

1. **MCQ**

Population inversion means:

- (a) More atoms in lower energy state
- (b) Equal atoms in all states
- (c) More atoms in excited state than ground state
- (d) No atoms in excited state

2. **True/False Type**

Statement 1: Optical resonator provides feedback in lasers.

Statement 2: Ordinary light is coherent.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. **MCQ**

Acceptance angle of an optical fiber is related to:

- (a) Mechanical strength
- (b) Light gathering ability
- (c) Thermal conductivity
- (d) Electrical resistance

### 6.2 Intermediate Level – Group Type

1. **MCQ**

Diode lasers are widely used in:

- (a) Welding only
- (b) Optical communication
- (c) Nuclear reactors
- (d) Steam engines

## 6.3 Advanced Level – Individual Type

### 1. Assertion and Reason

Assertion (A): Stimulated emission amplifies light intensity.

Reason (R): Emitted photons have same phase, frequency, and direction.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 2. Assertion and Reason

Assertion (A): Optical fibers experience very low signal loss.

Reason (R): Light is guided through repeated total internal reflections.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 6.4 Advanced Level – Group Type

### 1. Assertion and Reason

Assertion (A): Laser light is monochromatic.

Reason (R): Laser emission occurs due to transitions between fixed energy levels.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 7 Quantum Mechanics Principles – Set 1

### 7.1 Intermediate Level – Individual Type

1. MCQ

According to de Broglie hypothesis, matter particles exhibit:

- (a) Only particle nature
- (b) Only wave nature
- (c) Both wave and particle nature
- (d) Neither wave nor particle nature

2. True/False Type

Statement 1: It is impossible to simultaneously determine exact position and momentum.

Statement 2: This limitation is due to experimental defects only.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. MCQ

Schrodinger wave equation is used to determine:

- (a) Velocity only
- (b) Probability distribution
- (c) Electric field only
- (d) Current only

### 7.2 Intermediate Level – Group Type

1. MCQ

Quantum tunneling is the phenomenon where:

- (a) Particle loses energy completely
- (b) Particle crosses a barrier with finite probability
- (c) Electron stops moving
- (d) Wave disappears suddenly

### 7.3 Advanced Level – Individual Type

1. **Assertion and Reason**

Assertion (A): Electrons exhibit diffraction.

Reason (R): Electrons possess wave nature.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

2. **Assertion and Reason**

Assertion (A): Energy levels in an infinite potential well are quantized.

Reason (R): Electron wave function must satisfy boundary conditions.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 7.4 Advanced Level – Group Type

1. **Assertion and Reason**

Assertion (A): Scanning tunneling microscope works based on quantum tunneling.

Reason (R): Tunneling current depends on the distance between tip and surface.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 8 Quantum Mechanics Principles – Set 2

### 8.1 Intermediate Level – Individual Type

1. **MCQ**

The wavelength associated with a moving particle is called:

- (a) Compton wavelength
- (b) de Broglie wavelength
- (c) X-ray wavelength
- (d) Critical wavelength

2. **True/False Type**

Statement 1: Quantum dots exhibit size-dependent properties.

Statement 2: Quantum confinement occurs in quantum dots.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. **MCQ**

In a one-dimensional infinite potential well, the particle energy is:

- (a) Continuous
- (b) Infinite
- (c) Quantized
- (d) Negative always

### 8.2 Intermediate Level – Group Type

1. **MCQ**

Tunneling diode works based on:

- (a) Diffraction
- (b) Interference
- (c) Quantum tunneling
- (d) Polarization

### 8.3 Advanced Level – Individual Type

#### 1. Assertion and Reason

Assertion (A): de Broglie wavelength decreases with increase in momentum.

Reason (R): de Broglie wavelength is inversely proportional to momentum.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

#### 2. Assertion and Reason

Assertion (A): Quantum tunneling is possible even when particle energy is less than barrier energy.

Reason (R): Wave function penetrates into the barrier region.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 8.4 Advanced Level – Group Type

#### 1. Assertion and Reason

Assertion (A): Energy of a particle in an infinite potential well cannot be zero.

Reason (R): Zero energy would violate uncertainty principle.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 9 Quantum Band Theory – Set 1

### 9.1 Intermediate Level – Individual Type

1. **MCQ**

The forbidden energy gap in conductors is:

- (a) Very large
- (b) Zero
- (c) Equal to 5 eV
- (d) Infinite

2. **True/False Type**

Statement 1: Intrinsic semiconductors have equal electron and hole concentrations.

Statement 2: Doping converts intrinsic semiconductor into extrinsic semiconductor.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

3. **MCQ**

In n-type semiconductor, Fermi level shifts towards:

- (a) Valence band
- (b) Conduction band
- (c) Middle of band gap
- (d) Forbidden band only

### 9.2 Intermediate Level – Group Type

1. **MCQ**

Insulators have:

- (a) Small band gap
- (b) No band gap
- (c) Large band gap
- (d) Overlapping bands

### 9.3 Advanced Level – Individual Type

#### 1. Assertion and Reason

Assertion (A): Conductors have high electrical conductivity.

Reason (R): Valence and conduction bands overlap in conductors.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

#### 2. Assertion and Reason

Assertion (A): Doping increases conductivity of semiconductors.

Reason (R): Doping increases charge carrier concentration.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 9.4 Advanced Level – Group Type

#### 1. Assertion and Reason

Assertion (A): Fermi level lies near conduction band in n-type semiconductors.

Reason (R): Donor impurities contribute extra electrons.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## 10 Quantum Band Theory – Set 2

### 10.1 Intermediate Level – Individual Type

#### 1. MCQ

Semiconductors have band gap of the order of:

- (a) 0 eV
- (b) Very high value
- (c) About 1 eV
- (d) Infinite

#### 2. True/False Type

Statement 1: p-type semiconductors contain holes as majority carriers.

Statement 2: Acceptor impurities create holes in semiconductors.

- (a) Both statements are true
- (b) Both statements are false
- (c) Statement 1 is true and Statement 2 is false
- (d) Statement 1 is false and Statement 2 is true

#### 3. MCQ

The E-k diagram mainly represents relation between:

- (a) Energy and temperature
- (b) Energy and wave vector
- (c) Voltage and current
- (d) Pressure and volume

### 10.2 Intermediate Level – Group Type

#### 1. MCQ

In intrinsic semiconductors, conductivity increases with temperature because:

- (a) Carrier concentration increases
- (b) Band gap increases
- (c) Electrons disappear
- (d) Resistance becomes zero

### 10.3 Advanced Level – Individual Type

#### 1. Assertion and Reason

Assertion (A): Insulators do not conduct electricity under normal conditions.

Reason (R): Insulators possess a large forbidden energy gap.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

#### 2. Assertion and Reason

Assertion (A): Fermi level lies near valence band in p-type semiconductors.

Reason (R): Acceptor impurities create additional holes.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

### 10.4 Advanced Level – Group Type

#### 1. Assertion and Reason

Assertion (A): Semiconductors can behave as insulators at very low temperatures.

Reason (R): Very few charge carriers are available at low temperatures.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## **11 Answer Key**

### **11.1 Quantum Free Electron Theory – Set 1**

1. (c)
2. (a)
3. (b)
4. (b)
5. (a)
6. (a)
7. (a)

### **11.2 Quantum Free Electron Theory – Set 2**

1. (c)
2. (c)
3. (c)
4. (c)
5. (a)
6. (c)
7. (a)

### **11.3 Opto-electronics – Set 1**

1. (b)
2. (c)
3. (c)
4. (c)
5. (a)
6. (a)
7. (a)

### **11.4 Opto-electronics – Set 2**

1. (d)
2. (a)
3. (b)
4. (a)
5. (a)
6. (a)
7. (a)

### **11.5 Optical Fibers and Lasers – Set 1**

1. (b)
2. (a)
3. (c)
4. (b)
5. (a)
6. (a)
7. (a)

### **11.6 Optical Fibers and Lasers – Set 2**

1. (c)
2. (c)
3. (b)
4. (b)
5. (a)
6. (a)
7. (a)

### **11.7 Quantum Mechanics Principles – Set 1**

1. (c)
2. (c)
3. (b)
4. (b)
5. (a)
6. (a)
7. (a)

### **11.8 Quantum Mechanics Principles – Set 2**

1. (b)
2. (a)
3. (c)
4. (c)
5. (a)
6. (a)
7. (a)

### **11.9 Quantum Band Theory – Set 1**

1. (b)
2. (a)
3. (b)
4. (c)
5. (a)
6. (a)
7. (a)

### 11.10 Quantum Band Theory – Set 2

1. (c)
2. (a)
3. (b)
4. (a)
5. (a)
6. (a)
7. (a)

*End of Viva*